Eric Yang

eric.yang2@ucsf.edu | (310) 461-5746 | San Francisco, CA | Portfolio | Github | LinkedIn

EDUCATION

University of California, San Francisco (UCSF) San Francisco, CA
M.S. in Health Data Science Exp. Jun. 2026

University of California, Los Angeles (UCLA) Los Angeles, CA
B.S. in Computational and Systems Biology, Concentration in Biological Data Sciences Jun. 2024

- Departmental Highest Honors

RESEARCH EXPERIENCE

UCSF Department of Medicine San Francisco, CA
Data Science Specialist Apr. 2025 – Present
AI-Driven Feedback for Medical Student Clinical Assessment (OSCE Study)

- Lead development of a large language model (LLM) API system for clinical performance assessment, transforming OSCE audio/video encounters into structured scoring and faculty-aligned narrative feedback
- Architect and implement automated transcript-processing pipelines to enable scalable ingestion, analysis, and evaluation of clinical data
- Design and execute evaluation framework for model reliability and fairness, applying Cohen's kappa, semantic similarity, and natural language inference (NLI) to quantify agreement with human graders
- Direct model validation in collaboration with clinical faculty and refine outputs to align with expert grading standards and educational objectives
- Prepared and submitted IRB protocols, amendments, and continuing reviews for an IRB-approved study and grant proposal submissions, securing UCSF research funding of $4,822

UCSF Hswen Lab San Francisco, CA
Graduate Student Researcher Oct. 2024 – Present
Deep Learning for Alzheimer's Prediction: Ablation and Benchmarking

- Develop deep learning pipelines using convolutional neural networks (CNNs) and Vision Transformers to detect early Alzheimer's Disease and Related Dementias from 180K UK Biobank retinal images
- Design preprocessing workflows for retinal, macular, and vascular segmentation to enable structured feature extraction from fundus imaging data
- Construct multimodal predictive models integrating imaging and structured clinical data; benchmark performance against classical baselines (e.g., logistic regression) through structured evaluation metric

Behavioral and Systematic Analysis of Nicotine Pouch Use in Reddit

- Conduct large-scale NLP analysis of 38K+ Reddit posts to characterize sentiment, emotional framing, and behavioral patterns related to nicotine pouch use
- Apply computational methods including lemmatization, VADER sentiment analysis, and LDA topic modeling to identify recurring themes and public health-relevant discourse patterns

UCLA & PITTAN Los Angeles, CA
Undergraduate Thesis Jun. 2023 – Jun. 2024
Sweat Amino Acid Analysis for Non-Invasive Skin Assessment

- Investigated sweat amino acid composition as a non-invasive biomarker for dermatological health using fluorescence-based measurement and clinical comparison
- Analyzed 500+ clinical samples collected at Kobe University and benchmarked predictive performance against hospital biopsy standards
- Developed machine learning models, including random forests and convolutional neural networks (CNNs), to predict skin conditions and product recommendations from biochemical profiles
- Evaluated the translational potential of non-invasive biomarker approaches by assessing accuracy, cost, and clinical feasibility

UCLA Suthana Lab Santa Monica, CA
Research Assistant Mar. 2023 – Jun. 2024

- Analyzed 1,000+ EEG datasets using MATLAB to characterize neural signal patterns and assess data quality across experimental conditions
- Developed automated machine learning pipelines to improve efficiency, consistency, and reproducibility in neural signal processing workflows
- Supported IRB-approved clinical research workflows, including development of consent materials, patient coordination, and integration with ongoing study protocols

Institute of Neuro Innovation Santa Monica, CA
Data Analyst Aug. 2022 – Feb. 2023

- Supported clinical trials with 500+ patients testing VR audio therapy interventions for depression.
- Designed and tested three sound wavefront therapy protocols; applied Poisson distribution to classify treatment effectiveness; data entry and data analysis on excel

WORK EXPERIENCE

BOLERO Tokyo, Japan
Lead Data Scientist Apr. 2025 – Mar. 2026

- Lead design and deployment of AI-driven music analysis web secure systems across 100K+ multilingual tracks, modeling audio structure and artist-genre relationships
- Develop scalable machine learning pipelines for audio feature extraction using dimensionality reduction, clustering, and signal processing techniques (PCA, k-means, Fourier transforms)
- Build predictive models for engagement, recommendation, and content optimization using behavioral and time-series data
- Drive technical direction for product development and early-stage strategy, contributing to successful seed funding of ¥20M

PITTAN Kobe, Japan
Chief Technology Officer Jun. 2023 – Sep. 2023

- Developed Python-based systems for Raspberry Pi in digital convenience on platforms to enable biochemical data acquisition, analysis, and workflow automation for sweat-based diagnostics
- Led development of machine learning models for skin-condition inference using multimodal biological inputs (amino acids, moisture, lipid profiles)
- Directed end-to-end deployment of the PITTAN mobile application, integrating machine learning outputs, product design, and cloud-based infrastructure

ABSTRACTS AND PRESENTATIONS

Yang, E.; Vashisht, R.; La Joie, R.; Hoffmann, T.; Buto, P.; and Hswen, Y. (2026). "Limits of Fundus-Based Deep Learning for Alzheimer's Disease Prediction: Anatomical Ablation and Clinical Benchmarking in UK Biobank." Alzheimer's Association International Conference (AAIC 2026), London, United Kingdom; July 2026.

Yang, E. and Wong, F. (2026). "Simulating the Expert Evaluator: Immediate AI-Driven Feedback for Medical Student Oral Presentations and Written Documentation." UCSF Artificial Intelligence (AI) and Education Symposium, San Francisco, CA; Feb. 2026.

HONORS AND AWARDS

AAIC Conference Fellowship 2026

Skills & Interests

- **Languages:** Mandarin Chinese (native), Japanese (intermediate)
- **Programming:** C, C#, C++, HTML/CSS, Java, JavaScript, MATLAB, Python, R, SQL
- **Technical:** AI/ML, Clinical Design, Data Analysis, Retinal Imaging
- **Interests:** Percussion, Vocals, Digital Illustration, Creative Writing